

Bank Fraud Monitoring

A transaction-monitoring dashboard concept for surfacing high-value and cross-border activity for analyst review.

Status: dashboard and synthetic sample review documented

Executive summary

Built a Splunk dashboard that surfaces transaction records matching simple risk indicators, including high transaction amount and a country outside the expected baseline. The sample view highlighted a synthetic e-commerce transaction for CAD 4,875 associated with a Nigeria country code. This is an investigation lead, not a fraud verdict.

Scope and method

Fictional bank scenario; synthetic transaction records; Splunk search and dashboard configuration. No bank system, customer account, card network, or payment rail was accessed.

Results

Indicator / observation	Analyst interpretation
High-value transaction	CAD 4,875 record surfaced by the configured amount condition; verify customer history, approval context, and settlement status.
Cross-border signal	Country code differed from the assumed baseline; validate travel, merchant, and cardholder context before escalation.
Dashboard output	Tabular transaction review used to prioritize records; no automatic block, case creation, or fraud classification.

Evidence screenshots

Figure 1. Synthetic transaction dataset generation

```
akachi@akachi-ThinkPad-T440s:~$ mkdir -p ~/bank-fraud-splunk && python3 - <<'PY'
import csv, random
from datetime import datetime, timedelta

random.seed(42)
start = datetime(2026, 10, 4, 12, 0, 0)
rows = []
merchants = ["Grocery", "Fuel", "Pharmacy", "Electronics", "Online Retail"]

for i in range(80):
    rows.append({
        "event_time": (start + timedelta(minutes=random.randrange(120))).strftime("%Y-%m-%d %H:%M:%S"),
        "transaction_id": f"TX{i+1:04}",
        "card_id": f"CARD{random.randrange(1, 16):03}",
        "amount": f"{random.uniform(8, 220):.2f}",
        "merchant_category": random.choice(merchants),
        "country": "CA",
        "channel": random.choice(["POS", "ECOM"]),
        "auth_result": "APPROVED"
    })

for i in range(8):
    PYint("Created 91 synthetic transactions: ~/bank-fraud-splunk/transactions.csv")
Created 91 synthetic transactions: ~/bank-fraud-splunk/transactions.csv
akachi@akachi-ThinkPad-T440s:~$
```

Python script (fixed seed 42) generating baseline Canadian card transactions plus injected test records; 91 synthetic rows written to transactions.csv. Terminal capture, 17:33.

Figure 2. High-value and cross-border search

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```
source="transactions.csv"
| eval amount=tonumber(amount)
| where amount >= 1000 OR country!="CA"
| table event_time transaction_id card_id amount merchant_category country channel auth_result
```

The search results show 2 events. The table below displays the data for these events:

event_time	transaction_id	card_id	amount	merchant_category	country	channel	auth_result
2026-10-04 14:20:00	HIGH001	CARD901	4875.00	Electronics	NG	ECOM	APPROVED
2026-10-04 14:28:00	TRAVEL02	CARD902	690.00	Electronics	US	ECOM	APPROVED

amount >= 1000 OR country != "CA" returns two records: HIGH001 (4,875.00, NG, Electronics, ECOM) and TRAVEL02 (690.00, US, Electronics, ECOM). Both are investigation leads, not fraud verdicts. Splunk capture, 17:39.

Evidence screenshots (continued)

Figure 3. Repeated-decline (card testing) search

The screenshot shows the Splunk Enterprise search interface. The search bar contains the following query:

```
source="transactions.csv"
| stats count as attempts count(eval(auth_result="DECLINED")) as declines by card_id
| where attempts >= 6 AND declines >= 4
```

The search results show 91 events. The table below displays the results for card_id, attempts, and declines.

card_id	attempts	declines
CARD900	8	6

A second rule flags cards with at least 6 attempts and 4 declines. CARD900 is returned with 8 attempts and 6 declines, a pattern that can indicate card testing and needs cardholder and merchant context. Splunk capture, 17:39.

Figure 4. Bank Fraud Monitoring dashboard

The screenshot shows the Splunk Enterprise Bank Fraud Monitoring dashboard. The dashboard includes a table for High-Value and Cross-Border Transactions and a summary table for card_id, attempts, and declines.

event_time	transaction_id	card_id	amount	merchant_category	country	channel	auth_result
2026-10-04 14:20:00	HIGH001	CARD901	4875.00	Electronics	NG	ECOM	APPROVED
2026-10-04 14:28:00	TRAVEL02	CARD902	690.00	Electronics	US	ECOM	APPROVED

card_id	attempts	declines
CARD900	8	6

Dashboard combining the High-Value and Cross-Border Transactions panel with the repeated-decline panel. Tabular review only; no automatic block or case creation. Splunk capture, 18:05.

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Recommended follow-up

1. Confirm the transaction and cardholder identity using approved internal procedures.
2. Check device, merchant category, velocity, geography, prior disputes, and authentication signals.
3. Review authorization and settlement state with the payments team; use existing hold or recovery controls only when authorized.
4. Tune amount and geography rules against a representative baseline and measure false positives before operational use.

Limitations

The sample is synthetic and small. The dashboard uses simple indicators and does not establish intent, customer identity, loss, or settled fraud. A geography mismatch and high amount can both be legitimate. No production tuning or measurable detection rate is claimed.

Evidence basis

Prior Splunk dashboard screenshots and sample records discussed during project setup. Treat the screenshots as a visual demonstration; no independent production data was used.

Skills demonstrated

Splunk SPL, dashboard design, transaction triage, evidence-based escalation, and false-positive awareness.

Portfolio integrity note: This report describes an educational demonstration. It is not a client engagement, production security assessment, or certification of control effectiveness.